
Integrable Systems — Exercise Sheet 20

Generalised hydrodynamics, and the quantum Newton's cradle

Variant: Questions only

Conventions, as in Lecture 20. Hard rods: length a , unit mass, rapidity λ the velocity, $p_0 = \lambda$, $\varepsilon_0 = \lambda^2/2$; a state is a density $\rho(\lambda)$ with $n = \int \rho$, $J = \int \lambda \rho$, and $an < 1$. Dressing, as in Lecture 17: $f^{\text{dr}} = f + \frac{1}{2\pi} \int K(\cdot - \mu) \vartheta(\mu) f^{\text{dr}}(\mu) d\mu$ with $\vartheta = \rho/\rho_t$ and $2\pi\rho_t = (p'_0)^{\text{dr}}$; $v^{\text{eff}} = (\varepsilon'_0)^{\text{dr}}/(k'_0)^{\text{dr}}$; generalised hydrodynamics $\partial_t \rho + \partial_x (v^{\text{eff}} \rho) = 0$ for every λ . The Lieb–Liniger gas is in units $\hbar = 2m = 1$, with $\varepsilon_0 = k^2$ and $K(k) = 2c/(k^2 + c^2)$.

Exercise 1 (hard rods). (i) From Lecture 20's rate equation $v(\lambda) = \lambda + a \int \rho(\alpha) [v(\lambda) - v(\alpha)] d\alpha$, show that $\int \rho v = J$, and deduce $v^{\text{eff}} = (\lambda - aJ)/(1 - an)$. (ii) Show that dressing with the constant kernel $K = -a$ gives $(1)^{\text{dr}} = 1 - an$ and $(\lambda)^{\text{dr}} = \lambda - aJ$, and so the same v^{eff} . (iii) Boost every rod by u . Show that v^{eff} is boosted by u too.

Exercise 2 (the gas at large c is rods of negative length). (i) Show that $K(k) = 2/c + O(k^2/c^3)$. (ii) Dress the Lieb–Liniger gas with the constant kernel $K_0 = 2/c$ and show that $v^{\text{eff}} = 2(k - aJ)/(1 - an)$ with $a = -2/c$: hard rods of negative length, as Lecture 17's large- c limit says. (iii) Show that a local kernel $K = \kappa \delta$ would multiply every f by the same factor wherever $\kappa \vartheta \neq 2\pi$, so that v^{eff} would not be dressed at all.

Exercise 3 (which ratio is the velocity). Lecture 20's generalised TBA is $\eta(\lambda) = w(\lambda) - \frac{1}{2\pi} \int K(\lambda - \mu) \ln(1 + e^{-\eta(\mu)}) d\mu$, $\vartheta = 1/(1 + e^\eta)$. (i) Show that $w = (\varepsilon_0 - \mu)/T$ gives Lecture 17's Yang–Yang equation, with $\eta = \varepsilon/T$. (ii) Show that $\eta' = (w')^{\text{dr}}$, assuming $K \rightarrow 0$ at infinity and $\ln(1 + e^{-\eta})$ bounded. (iii) For free fermions ($K = 0$) with $\varepsilon_0 = \lambda^2$ and $w = \beta_2 \lambda^2 + \beta_4 \lambda^4$, compare v^{eff} with $\eta'/(2\pi\rho_t)$.

Exercise 4 (the cradle's free limits). In the ideal and the impenetrable limits $v^{\text{eff}} = 2\lambda$. In a harmonic trap Lecture 20's characteristics are $\dot{x} = 2\lambda$, $\dot{\lambda} = -\partial_x V$. (i) Why is $V = \omega^2 x^2/4$ the harmonic trap of frequency ω in these units? Solve the characteristics, and show that they rotate the plane $(x, 2\lambda/\omega)$ rigidly with period $2\pi/\omega$. (ii) Deduce that the filling returns exactly after each period. Show that at a quarter period positions and rescaled rapidities exchange roles, so that the momentum distribution $M_t(\lambda) = \int \rho dx$ is then $M_{\pi/2\omega}(\lambda) = \frac{2}{\omega} n_0(-2\lambda/\omega)$, with $n_0 = \int \rho d\lambda$ at $t = 0$. Is M_t constant for every initial state?

Exercise 5 (a lump of rods splitting). (*Computational.*) Take hard rods with $a = 1$ and the initial state $\rho_0(\lambda, x) = n_0(x) p(\lambda)$, $n_0(x) = 0.1(1 + 3e^{-(x/6)^2})$, with p the normalised sum of two Gaussians of width 0.45 centred at $\lambda = \pm 2$: the state of Lecture 20's figure. The background of density 0.1 fills the whole line; the lump is the excess over it. (i) Solve generalised hydrodynamics by the exact method of Exercise 7*. Plot $n(x, t)$ at $t = 0, 4, 8, 12$, and the free ($a = 0$) $n(x, 12)$, and locate the right-moving peak of $n(x, 12)$ for $a = 1$ and for $a = 0$. (ii) Solve it again by an upwind scheme in flux form on a periodic grid. Check that the scheme conserves the rods of each rapidity slice, and compare the two solutions at $t = 12$. (iii) Why does the interacting lump run ahead?

Exercise 6 (the Dictionary). The course's closing outcome. For each Part I entry below, give its Part II counterpart, the lecture where it lives, and the type of the correspondence: **T** a

theorem of this course (an explicit proved map or $\hbar \rightarrow 0$ limit), **A** a structural analogy, **C** cited. Poisson bracket; n integrals in involution; Poincaré sections; the linear action–angle system; $\oint p \cdot dq = 2\pi\hbar n$; permuted momenta; the classical displacement; the Lax matrix; the Sklyanin bracket; $r(\lambda)$; $\{t(\lambda), t(\mu)\} = 0$; $\text{tr } L^k$; the classical Yang–Baxter equation; the spectral curve; bands and gaps; the frequency $\partial K/\partial I$; the measure $dI d\varphi$; Calogero–Moser by reduction, and isochrony; the roots of $\mathfrak{sl}_N$; elastic non-diffractive scattering; invariant tori; KAM.

Exercise 7 ★. *Hard rods are contracted free particles.* Let point particles stream freely in a coordinate y : $\partial_t \hat{\rho} + \lambda \partial_y \hat{\rho} = 0$, with $\hat{\rho} \geq 0$ smooth, and $\hat{n} = \int \hat{\rho} d\lambda$, $\hat{j} = \int \lambda \hat{\rho} d\lambda$ finite and differentiable under the integral. No decay as $|y| \rightarrow \infty$ is assumed. (i) Show that $\hat{N}(y, t) = \int_0^y \hat{n}(y', t) dy' - \int_0^t \hat{j}(0, t') dt'$ has $\partial_y \hat{N} = \hat{n}$ and $\partial_t \hat{N} = -\hat{j}$. Give each particle a length $a > 0$ by the map $x = y + a\hat{N}(y, t)$, and let $\rho(\lambda, x, t) dx = \hat{\rho}(\lambda, y, t) dy$. (ii) Show that at each t the map is increasing and onto the line; that $\rho = \hat{\rho}/(1 + a\hat{n})$, $n = \hat{n}/(1 + a\hat{n})$, $j = \hat{j}/(1 + a\hat{n})$ and $1 - an = 1/(1 + a\hat{n})$; and that the point y at a fixed x moves as $\partial_t y|_x = a\hat{j}/(1 + a\hat{n})$. (iii) With $\hat{C}(\lambda, y, t) = \int_0^y \hat{\rho}(\lambda, y', t) dy' - \lambda \int_0^t \hat{\rho}(\lambda, 0, t') dt'$ and $C(\lambda, x, t) = \hat{C}(\lambda, y, t)$, show that $\partial_x C = \rho$ and $\partial_t C|_x = -\rho v^{\text{eff}}$ with $v^{\text{eff}} = (\lambda - aj)/(1 - an)$, and deduce that ρ solves generalised hydrodynamics for hard rods.